

Double-click (or enter) to edit

Employee Attrition Prediction using Machine Learning

Objective

Employee attrition is one of the biggest challenges faced by organizations because losing experienced employees increases recruitment costs, training expenses, and affects overall productivity.

The objective of this project is to build a machine learning model that predicts whether an employee is likely to leave the company based on factors such as job role, monthly income, work-life balance, years at the company, overtime, and job satisfaction.

The project also aims to identify the key factors responsible for employee attrition and provide actionable business recommendations that can help the HR department improve employee retention.

Business Problem

Employee attrition is a major concern for organizations because replacing skilled employees requires significant time and cost. HR teams need a proactive way to identify employees who are at risk of leaving.

Using historical employee data, this project develops predictive models that help HR identify high-risk employees and understand the factors contributing to attrition. These insights can support better retention strategies and workforce planning.

```
# =====
# Import Required Libraries
# =====

import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.ensemble import GradientBoostingClassifier

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    roc_auc_score,
    confusion_matrix,
    classification_report,
    RocCurveDisplay
)

import warnings
warnings.filterwarnings("ignore")
```

Step 1: Data Loading and Exploration

The first step is to load the employee attrition dataset and understand its structure. We will examine the number of records, available features, target variable, and basic information before performing any preprocessing.

```
# Load Dataset

df = pd.read_csv("WA_Fn-UseC_-HR-Employee-Attrition.csv")

print("Dataset Loaded Successfully!")

print("\nDataset Shape:")
print(df.shape)

print("\nFirst 10 Rows")
display(df.head(10))
```

Dataset Loaded Successfully!																		
Dataset Shape: (1470, 35)																		
First 10 Rows																		
	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	...	RelationshipSatisfaction	StandardHours	StockOptionLevel	TotalWorkingYears	TrainingTimesLastYear	WorkLifeBalance	
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	...	1	80	0	8	0	1	
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	1	2	...	4	80	1	10	3	3	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	1	4	...	2	80	0	7	3	3	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	1	5	...	3	80	0	8	3	3	
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	7	...	4	80	1	6	3	3	
5	32	No	Travel_Frequently	1005	Research & Development	2	2	Life Sciences	1	8	...	3	80	0	8	2	2	
6	59	No	Travel_Rarely	1324	Research & Development	3	3	Medical	1	10	...	1	80	3	12	3	2	
7	30	No	Travel_Rarely	1358	Research & Development	24	1	Life Sciences	1	11	...	2	80	1	1	2	3	
8	38	No	Travel_Frequently	216	Research & Development	23	3	Life Sciences	1	12	...	2	80	0	10	2	3	
9	36	No	Travel_Rarely	1299	Research & Development	27	3	Medical	1	13	...	2	80	2	17	3	2	
10 rows × 35 columns																		

```
print("Column Names:\n")
print(df.columns.tolist())

print("\n")

print("Dataset Information:\n")
df.info()

print("\n")

print("Statistical Summary:\n")
display(df.describe())
```


Column Names:

['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department', 'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount', 'EmployeeNumber', 'EnvironmentSatisfaction', 'Gender', 'HourlyRate', 'JobInvolvement', 'JobLevel', 'JobRole', 'JobSa

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Age                                    1470 non-null   int64
1   Attrition                            1470 non-null   object
2   BusinessTravel                       1470 non-null   object
3   DailyRate                            1470 non-null   int64
4   Department                           1470 non-null   object
5   DistanceFromHome                    1470 non-null   int64
6   Education                            1470 non-null   int64
7   EducationField                       1470 non-null   object
8   EmployeeCount                       1470 non-null   int64
9   EmployeeNumber                      1470 non-null   int64
10  EnvironmentSatisfaction              1470 non-null   int64
11  Gender                               1470 non-null   object
12  HourlyRate                           1470 non-null   int64
13  JobInvolvement                      1470 non-null   int64
14  JobLevel                            1470 non-null   int64
15  JobRole                              1470 non-null   object
16  JobSatisfaction                     1470 non-null   int64
17  MaritalStatus                       1470 non-null   object
18  MonthlyIncome                       1470 non-null   int64
19  MonthlyRate                         1470 non-null   int64
20  NumCompaniesWorked                  1470 non-null   int64
21  Over18                              1470 non-null   object
22  OverTime                            1470 non-null   object
23  PercentSalaryHike                   1470 non-null   int64
24  PerformanceRating                   1470 non-null   int64
25  RelationshipSatisfaction              1470 non-null   int64
26  StandardHours                       1470 non-null   int64
27  StockOptionLevel                    1470 non-null   int64
28  TotalWorkingYears                   1470 non-null   int64
29  TrainingTimesLastYear               1470 non-null   int64
30  WorkLifeBalance                     1470 non-null   int64
31  YearsAtCompany                      1470 non-null   int64
32  YearsInCurrentRole                  1470 non-null   int64
33  YearsSinceLastPromotion              1470 non-null   int64
34  YearsWithCurrManager                1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

Statistical Summary:

	Age	DailyRate	DistanceFromHome	Education	EmployeeCount	EmployeeNumber	EnvironmentSatisfaction	HourlyRate	JobInvolvement	JobLevel	...	RelationshipSatisfaction	StandardHours	StockOptionLevel	TotalWorkingYears	TrainingTimesLas
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.0	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000	...	1470.000000	1470.0	1470.000000	1470.000000	1470.0
mean	36.923810	802.485714	9.192517	2.912925	1.0	1024.865306	2.721769	65.891156	2.729932	2.063946	...	2.712245	80.0	0.793878	11.279592	2.7
std	9.135373	403.509100	8.106864	1.024165	0.0	602.024335	1.093082	20.329428	0.711561	1.106940	...	1.081209	0.0	0.852077	7.780782	1.2
min	18.000000	102.000000	1.000000	1.000000	1.0	1.000000	1.000000	30.000000	1.000000	1.000000	...	1.000000	80.0	0.000000	0.000000	0.0
25%	30.000000	465.000000	2.000000	2.000000	1.0	491.250000	2.000000	48.000000	2.000000	1.000000	...	2.000000	80.0	0.000000	6.000000	2.0
50%	36.000000	802.000000	7.000000	3.000000	1.0	1020.500000	3.000000	66.000000	3.000000	2.000000	...	3.000000	80.0	1.000000	10.000000	3.0
75%	43.000000	1157.000000	14.000000	4.000000	1.0	1555.750000	4.000000	83.750000	3.000000	3.000000	...	4.000000	80.0	1.000000	15.000000	3.0
max	60.000000	1499.000000	29.000000	5.000000	1.0	2068.000000	4.000000	100.000000	4.000000	5.000000	...	4.000000	80.0	3.000000	40.000000	6.0

8 rows × 26 columns

Attrition Analysis

In this section, we analyze the target variable (Attrition) to understand how many employees stayed and left the company. We also calculate the attrition rate and examine whether the dataset is balanced or imbalanced before building machine learning models.

```
# =====
# Target Variable Analysis
# =====

# Count employees
attrition_counts = df['Attrition'].value_counts()

print("Employee Attrition Count")
print(attrition_counts)

print()

# Calculate attrition rate
attrition_rate = (attrition_counts['Yes'] / len(df)) * 100

print(f"Attrition Rate: {attrition_rate:.2f}%")

print()

# Numeric and categorical columns
numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns
categorical_cols = df.select_dtypes(include=['object']).columns

print("Number of Numerical Columns:", len(numeric_cols))
print("Number of Categorical Columns:", len(categorical_cols))

print()

# Dataset balance
if attrition_rate < 30:
    print("Observation: The dataset is imbalanced because employees leaving the company represent a much smaller portion of the dataset.")
else:
    print("Observation: The dataset is relatively balanced.")
```

Employee Attrition Count
Attrition
No 1233
Yes 237
Name: count, dtype: int64

Attrition Rate: 16.12%

Number of Numerical Columns: 26
Number of Categorical Columns: 9

Observation: The dataset is imbalanced because employees leaving the company represent a much smaller portion of the dataset.

Observation

The dataset contains significantly more employees who stayed than employees who left the company. The attrition rate is approximately 16%, indicating that the dataset is imbalanced. This imbalance is important because machine learning models may become biased toward predicting the majority class. Therefore, class balancing techniques such as `class_weight='balanced'` will be used during model training.

Step 2: Data Cleaning and Preprocessing

Before training machine learning models, the dataset must be cleaned and transformed into a suitable format. In this step, we will remove irrelevant columns, encode categorical variables, scale numerical features, and prepare the dataset for classification models.

```
# =====
# Check Missing Values
# =====

print("Missing Values in Each Column")
print(df.isnull().sum())

print("\nTotal Missing Values:", df.isnull().sum().sum())
```

Missing Values in Each Column
Age 0
Attrition 0
BusinessTravel 0
DailyRate 0
Department 0
DistanceFromHome 0
Education 0
EducationField 0
EmployeeCount 0
EmployeeNumber 0
EnvironmentSatisfaction 0
Gender 0
HourlyRate 0
JobInvolvement 0
JobLevel 0
JobRole 0
JobSatisfaction 0
MaritalStatus 0
MonthlyIncome 0
MonthlyRate 0
NumCompaniesWorked 0
Over18 0
OverTime 0
PercentSalaryHike 0
PerformanceRating 0
RelationshipSatisfaction 0
StandardHours 0
StockOptionLevel 0
TotalWorkingYears 0
TrainingTimesLastYear 0
WorkLifeBalance 0
YearsAtCompany 0


```
YearsInCurrentRole      0
YearsSinceLastPromotion  0
YearsWithCurrManager     0
dtype: int64

Total Missing Values: 0

# =====
# Remove Irrelevant Columns
# =====

columns_to_drop = [
    'EmployeeNumber',
    'EmployeeCount',
    'Over18',
    'StandardHours'
]

df = df.drop(columns=columns_to_drop)
print("Remaining Columns:", df.shape[1])

Remaining Columns: 31
```

Step 3: Encoding Categorical Variables

Machine learning models require numerical input. Therefore, all categorical features are converted into numerical values using Label Encoding. This ensures that the models can process the data effectively while preserving the information contained in categorical variables.

```
# =====
# Encode Categorical Variables
# =====

from sklearn.preprocessing import LabelEncoder

# Create LabelEncoder object
label_encoder = LabelEncoder()

# Find categorical columns
categorical_columns = df.select_dtypes(include='object').columns

print("Categorical Columns:")
print(categorical_columns)

# Apply Label Encoding
for column in categorical_columns:
    df[column] = label_encoder.fit_transform(df[column])

    print("\nEncoding Completed Successfully!")

    display(df.head())

Categorical Columns:
Index(['Attrition', 'BusinessTravel', 'Department', 'EducationField', 'Gender',
      'JobRole', 'MaritalStatus', 'OverTime'],
      dtype='object')

Encoding Completed Successfully!
   Age  Attrition  BusinessTravel  DailyRate  Department  DistanceFromHome  Education  EducationField  EnvironmentSatisfaction  Gender  ...  PerformanceRating  RelationshipSatisfaction  StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  WorkLifeB
0   41         1         Travel_Rarely      1102         Sales                1         2         Life Sciences                2  Female  ...              3              1              0              8              0
1   49         0      Travel_Frequently       279  Research & Development                8         1         Life Sciences                3   Male  ...              4              4              1              10              3
2   37         1         Travel_Rarely      1373  Research & Development                2         2              Other                4   Male  ...              3              2              0              7              3
3   33         0      Travel_Frequently      1392  Research & Development                3         4         Life Sciences                4  Female  ...              3              3              0              8              3
4   27         0         Travel_Rarely       591  Research & Development                2         1           Medical                1   Male  ...              3              4              1              6              3
5 rows x 31 columns

Encoding Completed Successfully!
   Age  Attrition  BusinessTravel  DailyRate  Department  DistanceFromHome  Education  EducationField  EnvironmentSatisfaction  Gender  ...  PerformanceRating  RelationshipSatisfaction  StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  WorkLifeB
0   41         1         2         1102         Sales                1         2         Life Sciences                2  Female  ...              3              1              0              8              0
1   49         0         1         279  Research & Development                8         1         Life Sciences                3   Male  ...              4              4              1              10              3
2   37         1         2      1373  Research & Development                2         2              Other                4   Male  ...              3              2              0              7              3
3   33         0         1      1392  Research & Development                3         4         Life Sciences                4  Female  ...              3              3              0              8              3
4   27         0         2         591  Research & Development                2         1           Medical                1   Male  ...              3              4              1              6              3
5 rows x 31 columns

Encoding Completed Successfully!
   Age  Attrition  BusinessTravel  DailyRate  Department  DistanceFromHome  Education  EducationField  EnvironmentSatisfaction  Gender  ...  PerformanceRating  RelationshipSatisfaction  StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  WorkLifeB
0   41         1         2      1102         2                1         2         Life Sciences                2  Female  ...              3              1              0              8              0
1   49         0         1         279         1                8         1         Life Sciences                3   Male  ...              4              4              1              10              3
2   37         1         2      1373         1                2         2              Other                4   Male  ...              3              2              0              7              3
3   33         0         1      1392         1                3         4         Life Sciences                4  Female  ...              3              3              0              8              3
4   27         0         2         591         1                2         1           Medical                1   Male  ...              3              4              1              6              3
5 rows x 31 columns

Encoding Completed Successfully!
   Age  Attrition  BusinessTravel  DailyRate  Department  DistanceFromHome  Education  EducationField  EnvironmentSatisfaction  Gender  ...  PerformanceRating  RelationshipSatisfaction  StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  WorkLifeB
0   41         1         2      1102         2                1         2              1                2  Female  ...              3              1              0              8              0
1   49         0         1         279         1                8         1              1                3   Male  ...              4              4              1              10              3
2   37         1         2      1373         1                2         2              4                4   Male  ...              3              2              0              7              3
3   33         0         1      1392         1                3         4              1                4  Female  ...              3              3              0              8              3
4   27         0         2         591         1                2         1              3                1   Male  ...              3              4              1              6              3
5 rows x 31 columns

Encoding Completed Successfully!
   Age  Attrition  BusinessTravel  DailyRate  Department  DistanceFromHome  Education  EducationField  EnvironmentSatisfaction  Gender  ...  PerformanceRating  RelationshipSatisfaction  StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  WorkLifeB
0   41         1         2      1102         2                1         2              1                2    0  ...              3              1              0              8              0
1   49         0         1         279         1                8         1              1                3    1  ...              4              4              1              10              3
2   37         1         2      1373         1                2         2              4                4    1  ...              3              2              0              7              3
3   33         0         1      1392         1                3         4              1                4    0  ...              3              3              0              8              3
4   27         0         2         591         1                2         1              3                1    1  ...              3              4              1              6              3
5 rows x 31 columns

Encoding Completed Successfully!
   Age  Attrition  BusinessTravel  DailyRate  Department  DistanceFromHome  Education  EducationField  EnvironmentSatisfaction  Gender  ...  PerformanceRating  RelationshipSatisfaction  StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  WorkLifeB
0   41         1         2      1102         2                1         2              1                2    0  ...              3              1              0              8              0
1   49         0         1         279         1                8         1              1                3    1  ...              4              4              1              10              3
2   37         1         2      1373         1                2         2              4                4    1  ...              3              2              0              7              3
3   33         0         1      1392         1                3         4              1                4    0  ...              3              3              0              8              3
4   27         0         2         591         1                2         1              3                1    1  ...              3              4              1              6              3
5 rows x 31 columns

Encoding Completed Successfully!
   Age  Attrition  BusinessTravel  DailyRate  Department  DistanceFromHome  Education  EducationField  EnvironmentSatisfaction  Gender  ...  PerformanceRating  RelationshipSatisfaction  StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  WorkLifeB
0   41         1         2      1102         2                1         2              1                2    0  ...              3              1              0              8              0
1   49         0         1         279         1                8         1              1                3    1  ...              4              4              1              10              3
2   37         1         2      1373         1                2         2              4                4    1  ...              3              2              0              7              3
3   33         0         1      1392         1                3         4              1                4    0  ...              3              3              0              8              3
4   27         0         2         591         1                2         1              3                1    1  ...              3              4              1              6              3
5 rows x 31 columns

Encoding Completed Successfully!
   Age  Attrition  BusinessTravel  DailyRate  Department  DistanceFromHome  Education  EducationField  EnvironmentSatisfaction  Gender  ...  PerformanceRating  RelationshipSatisfaction  StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  WorkLifeB
0   41         1         2      1102         2                1         2              1                2    0  ...              3              1              0              8              0
1   49         0         1         279         1                8         1              1                3    1  ...              4              4              1              10              3
2   37         1         2      1373         1                2         2              4                4    1  ...              3              2              0              7              3
3   33         0         1      1392         1                3         4              1                4    0  ...              3              3              0              8              3
4   27         0         2         591         1                2         1              3                1    1  ...              3              4              1              6              3
5 rows x 31 columns
```

Step 4: Feature Scaling and Train-Test Split

Before training machine learning models, the dataset is divided into training and testing sets. Feature scaling is then applied to numerical variables to ensure that features with larger values do not dominate the learning process. This improves the performance of many machine learning algorithms.

```
# =====
# Split Features and Target
# =====

X = df.drop("Attrition", axis=1)
y = df["Attrition"]

print("Feature Matrix Shape:", X.shape)
print("Target Shape:", y.shape)

Feature Matrix Shape: (1470, 30)
Target Shape: (1470,)
```

```
# =====
# Train-Test Split
# =====

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("Training Samples:", X_train.shape)
print("Testing Samples:", X_test.shape)

Training Samples: (1176, 30)
Testing Samples: (294, 30)
```

```
# =====
# Feature Scaling
# =====

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("Feature Scaling Completed Successfully!")

Feature Scaling Completed Successfully!
```

Step 5: Machine Learning Model Training

Three machine learning classification models are trained to predict employee attrition:

- 1. Logistic Regression
- 2. Random Forest Classifier
- 3. Gradient Boosting Classifier

The performance of each model will be evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC Score.

```
# =====
# Logistic Regression
# =====

log_model = LogisticRegression(
    random_state=42,
    class_weight='balanced',
    max_iter=1000
)

log_model.fit(X_train_scaled, y_train)
log_pred = log_model.predict(X_test_scaled)
log_prob = log_model.predict_proba(X_test_scaled)[:, 1]

print("Logistic Regression Model Trained Successfully!")

Logistic Regression Model Trained Successfully!
```

```
# =====
# Random Forest Classifier
# =====

rf_model = RandomForestClassifier(
    n_estimators=200,
    random_state=42,
    class_weight='balanced'
)

rf_model.fit(X_train, y_train)

rf_pred = rf_model.predict(X_test)
rf_prob = rf_model.predict_proba(X_test)[:, 1]

print("Random Forest Model Trained Successfully!")

Random Forest Model Trained Successfully!
```

```
# =====
# Gradient Boosting Classifier
# =====

gb_model = GradientBoostingClassifier(
    random_state=42
)

gb_model.fit(X_train, y_train)

gb_pred = gb_model.predict(X_test)
gb_prob = gb_model.predict_proba(X_test)[:, 1]

print("Gradient Boosting Model Trained Successfully!")

Gradient Boosting Model Trained Successfully!
```

Step 6: Model Evaluation

The trained models are evaluated using five performance metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

These metrics help compare the models and identify the best-performing algorithm for employee attrition prediction.

```
# =====
# Function to Evaluate Models
# =====

def evaluate_model(name, y_true, y_pred, y_prob):
    return {
        "Model": name,
        "Accuracy": accuracy_score(y_true, y_pred),
        "Precision": precision_score(y_true, y_pred),
        "Recall": recall_score(y_true, y_pred),
        "F1 Score": f1_score(y_true, y_pred),
        "ROC-AUC": roc_auc_score(y_true, y_prob)
    }
```

```
# =====
# Compare All Models
# =====

results = pd.DataFrame([

    evaluate_model(
        "Logistic Regression",
        y_test,
        log_pred,
        log_prob
    ),

    evaluate_model(
        "Random Forest",
        y_test,
        rf_pred,
        rf_prob
    ),

    evaluate_model(
        "Gradient Boosting",
        y_test,
        gb_pred,
        gb_prob
    )

])

results = results.round(4)
```



```
display(results)

# Best Model

best_model = results.loc[
    results["F1 Score"].idxmax()
]

print("Best Performing Model")
display(best_model)

Best Performing Model
0
Model    Logistic Regression
Accuracy    0.7517
Precision    0.3673
Recall    0.766
F1 Score    0.4966
ROC-AUC    0.8074

dtype: object
```

Step 7: Feature Importance Analysis

Feature importance helps identify the variables that have the greatest impact on employee attrition. Understanding these factors enables HR teams to design effective employee retention strategies.

```
# =====
# Feature Importance
# =====

feature_importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": rf_model.feature_importances_
})

feature_importance = feature_importance.sort_values(
    by="Importance",
    ascending=False
)

display(feature_importance.head(10))

Feature    Importance
15    MonthlyIncome    0.076928
0            Age    0.070421
2        DailyRate    0.056734
23    TotalWorkingYears    0.056145
26    YearsAtCompany    0.049168
16        MonthlyRate    0.047601
18            OverTime    0.047033
4    DistanceFromHome    0.046327
9        HourlyRate    0.046270
29    YearsWithCurrManager    0.040347

# Top 10 Important Features

plt.figure(figsize=(10,6))

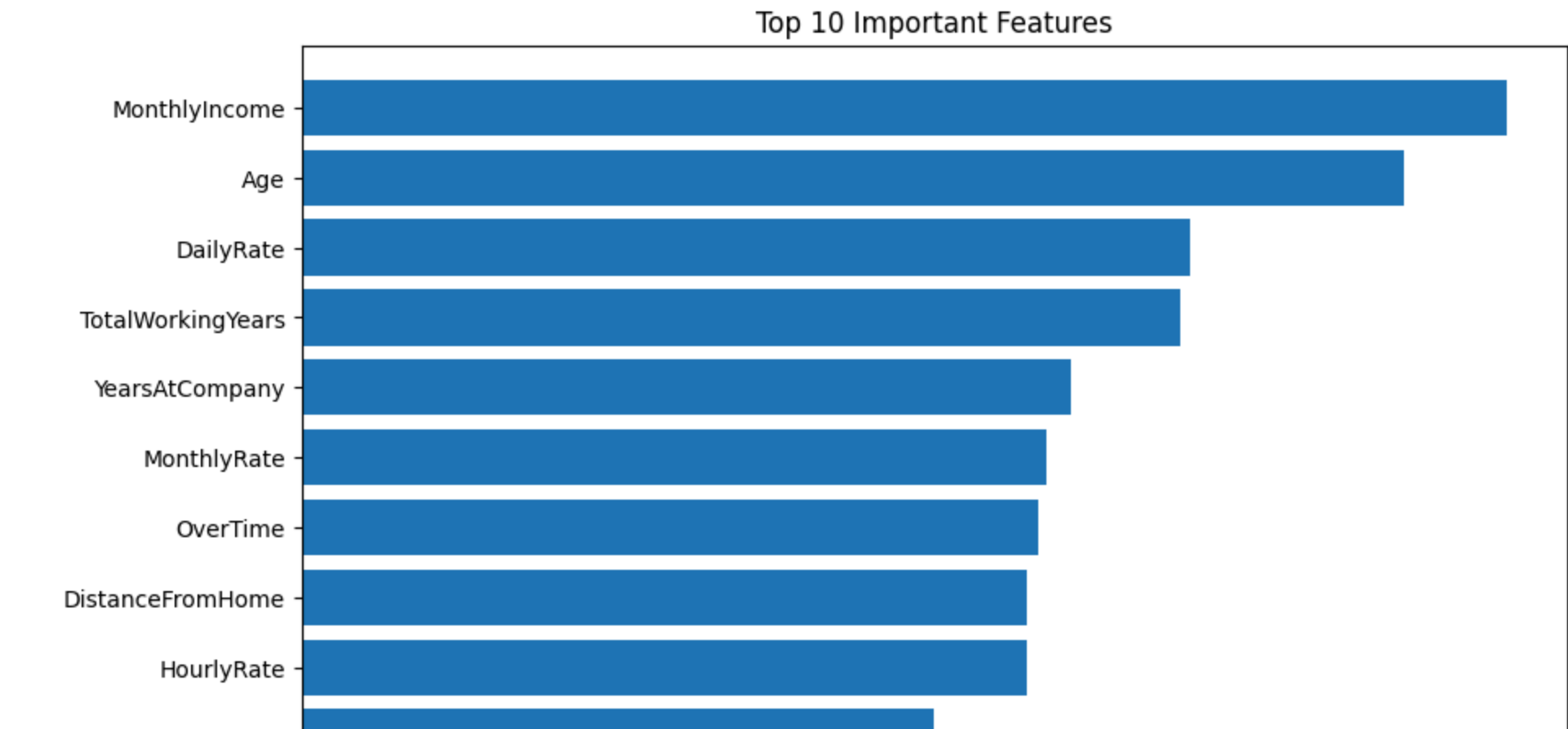
plt.barh(
    feature_importance["Feature"][:10],
    feature_importance["Importance"][:10]
)

plt.title("Top 10 Important Features")
plt.xlabel("Importance")

plt.gca().invert_yaxis()

# Save the figure
plt.savefig("feature_importance.png", dpi=300, bbox_inches="tight")

plt.show()
```



Step 8: Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) helps understand employee attrition patterns by analyzing relationships between employee characteristics and the target variable. These visualizations provide valuable business insights that can help HR teams identify high-risk groups and design effective retention strategies.

```
import os

os.makedirs("charts", exist_ok=True)

print("Charts folder created successfully!")

Charts folder created successfully!

plt.figure(figsize=(8,5))

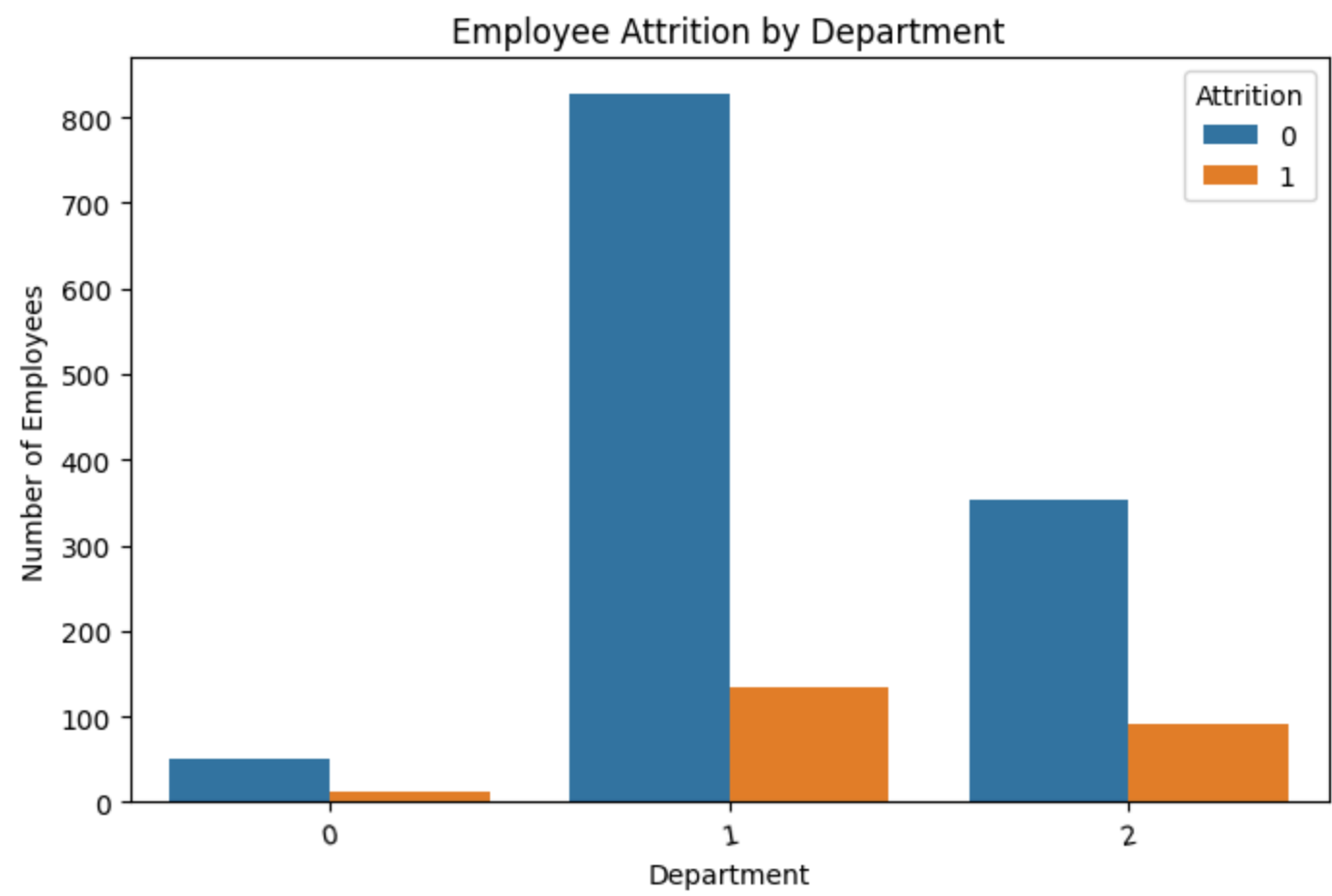
sns.countplot(
    data=df,
    x="Department",
    hue="Attrition"
)

plt.title("Employee Attrition by Department")
plt.xlabel("Department")
plt.ylabel("Number of Employees")

plt.xticks(rotation=10)

plt.savefig(
    "charts/attrition_by_department.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```

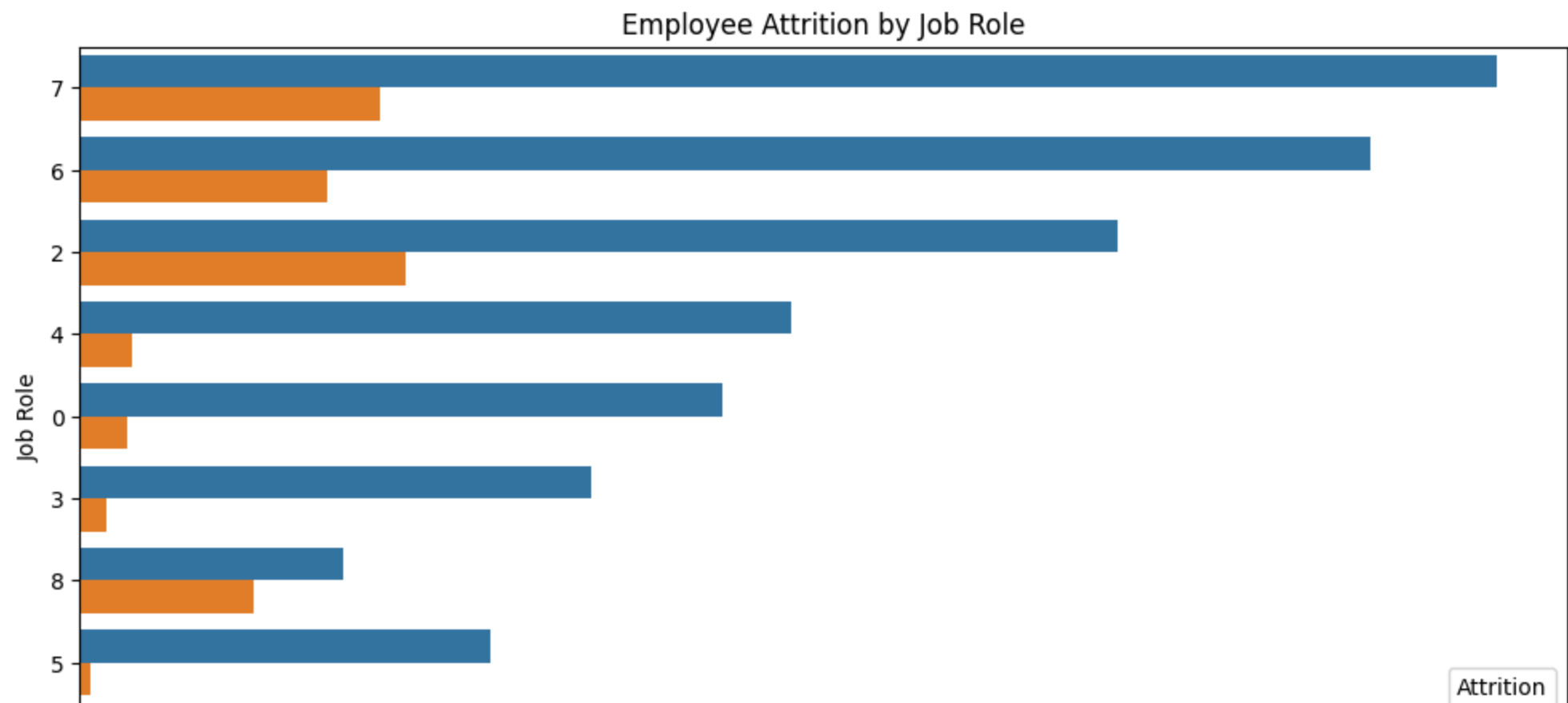



```
plt.figure(figsize=(12,6))

sns.countplot(
    data=df,
    y="JobRole",
    hue="Attrition",
    order=df["JobRole"].value_counts().index
)

plt.title("Employee Attrition by Job Role")
plt.xlabel("Number of Employees")
plt.ylabel("Job Role")
plt.savefig(
    "charts/attrition_by_jobrole.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



Monthly Income vs Employee Attrition

This visualization compares the monthly income distribution of employees who stayed versus those who left the company.

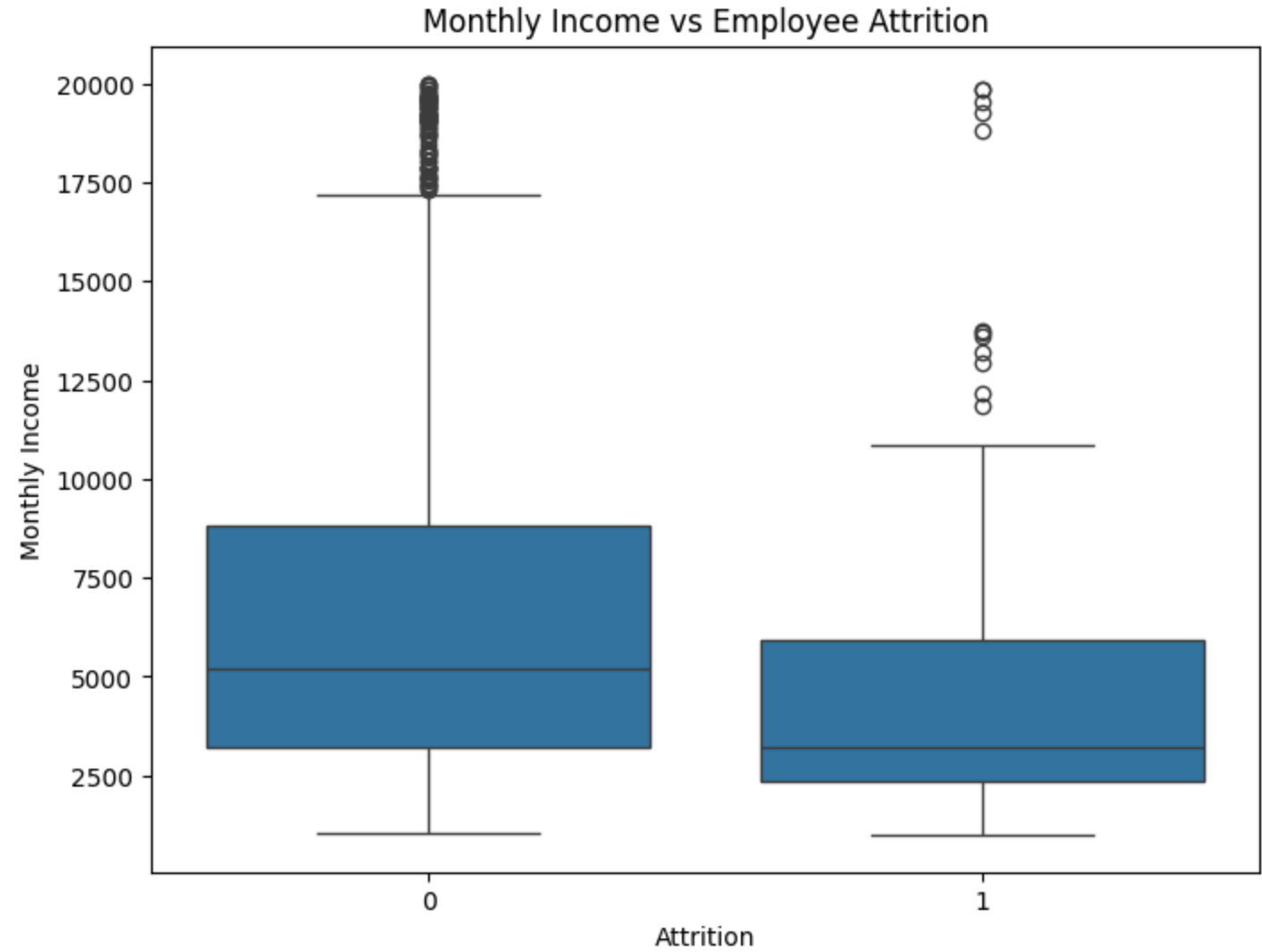
```
plt.figure(figsize=(8,6))

sns.boxplot(
    data=df,
    x="Attrition",
    y="MonthlyIncome"
)

plt.title("Monthly Income vs Employee Attrition")
plt.xlabel("Attrition")
plt.ylabel("Monthly Income")

plt.savefig(
    "charts/monthly_income_vs_attrition.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



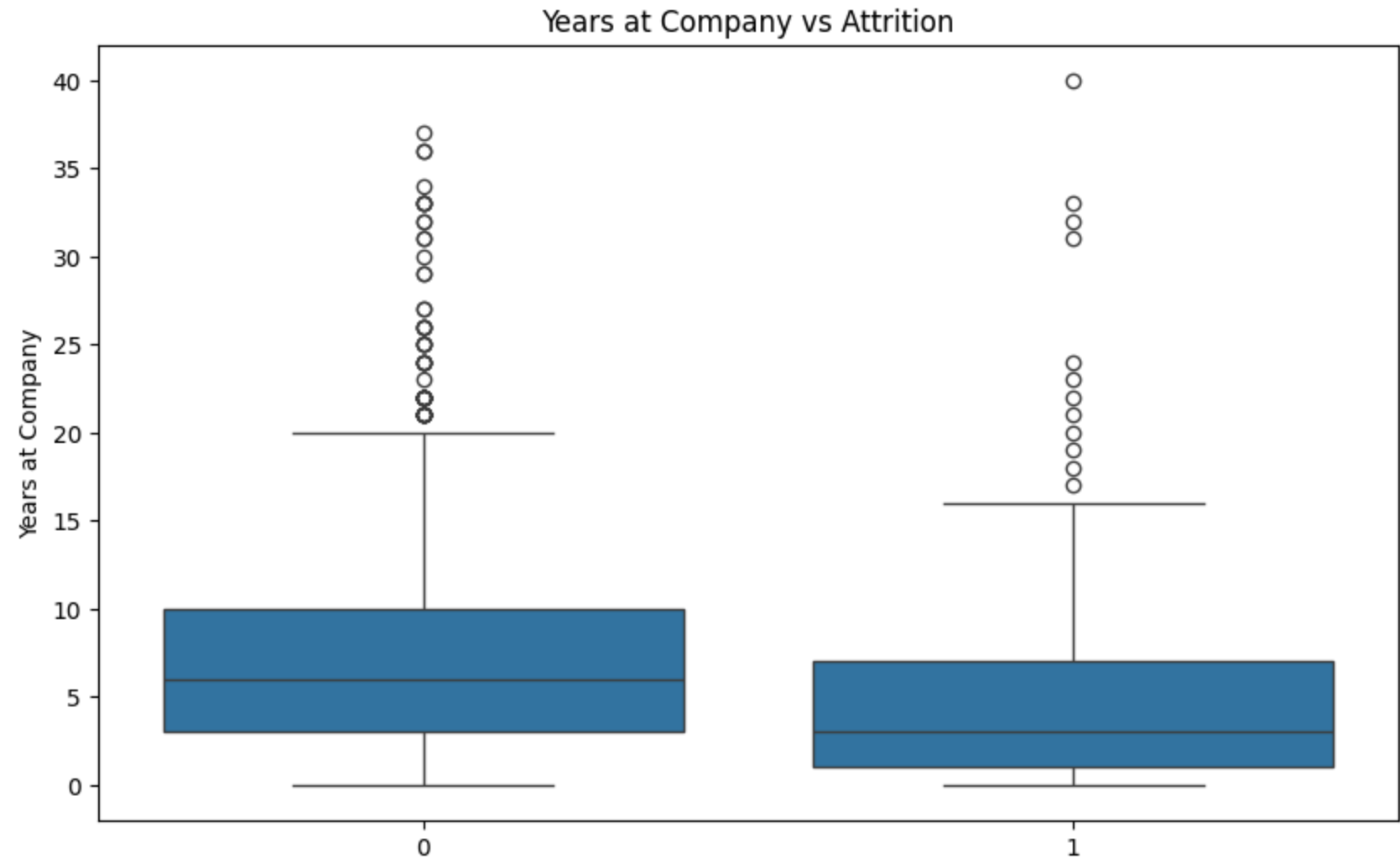
```
plt.figure(figsize=(10,6))

sns.boxplot(
    data=df,
    x="Attrition",
    y="YearsAtCompany"
)

plt.title("Years at Company vs Attrition")
plt.xlabel("Attrition")
plt.ylabel("Years at Company")

plt.savefig(
    "charts/years_at_company_vs_attrition.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



```
from sklearn.metrics import ConfusionMatrixDisplay

plt.figure(figsize=(6,6))
```

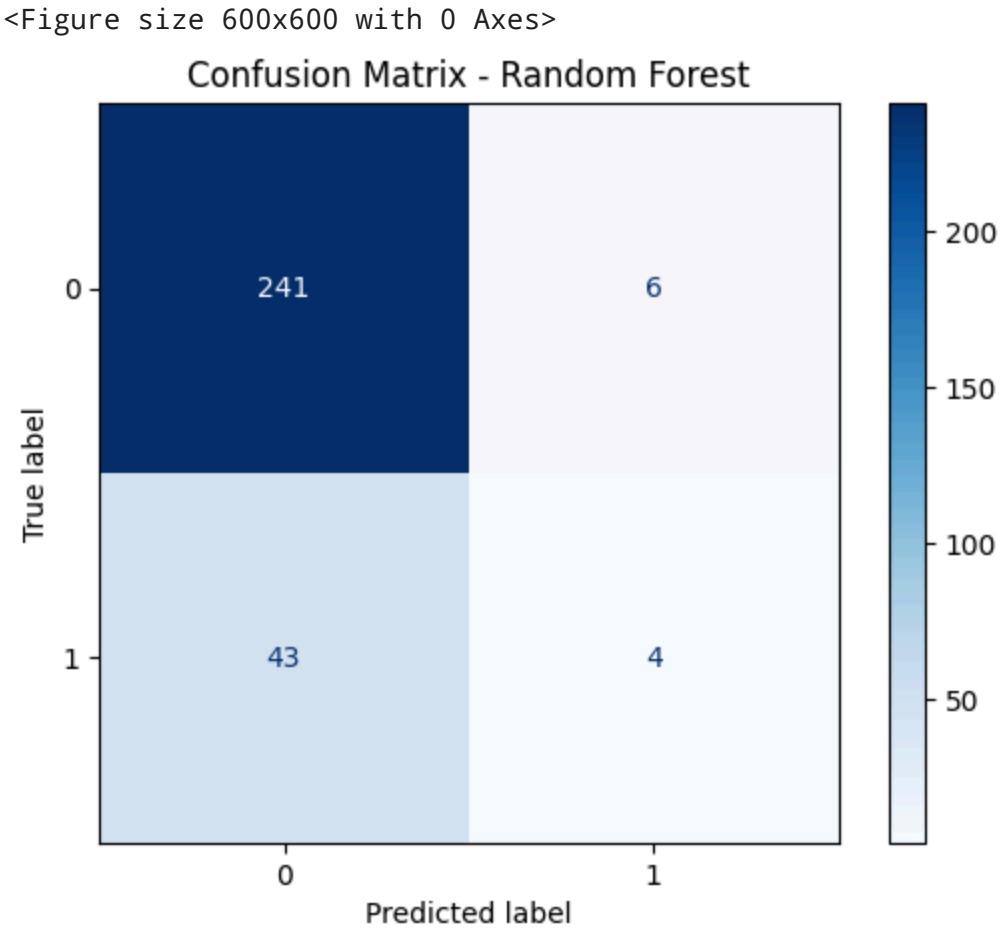


```
ConfusionMatrixDisplay.from_estimator(
    rf_model,
    X_test,
    y_test,
    cmap="Blues"
)

plt.title("Confusion Matrix - Random Forest")

plt.savefig(
    "charts/confusion_matrix.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



```
plt.figure(figsize=(8,6))

RocCurveDisplay.from_predictions(
    y_test,
    log_prob,
    name="Logistic Regression"
)

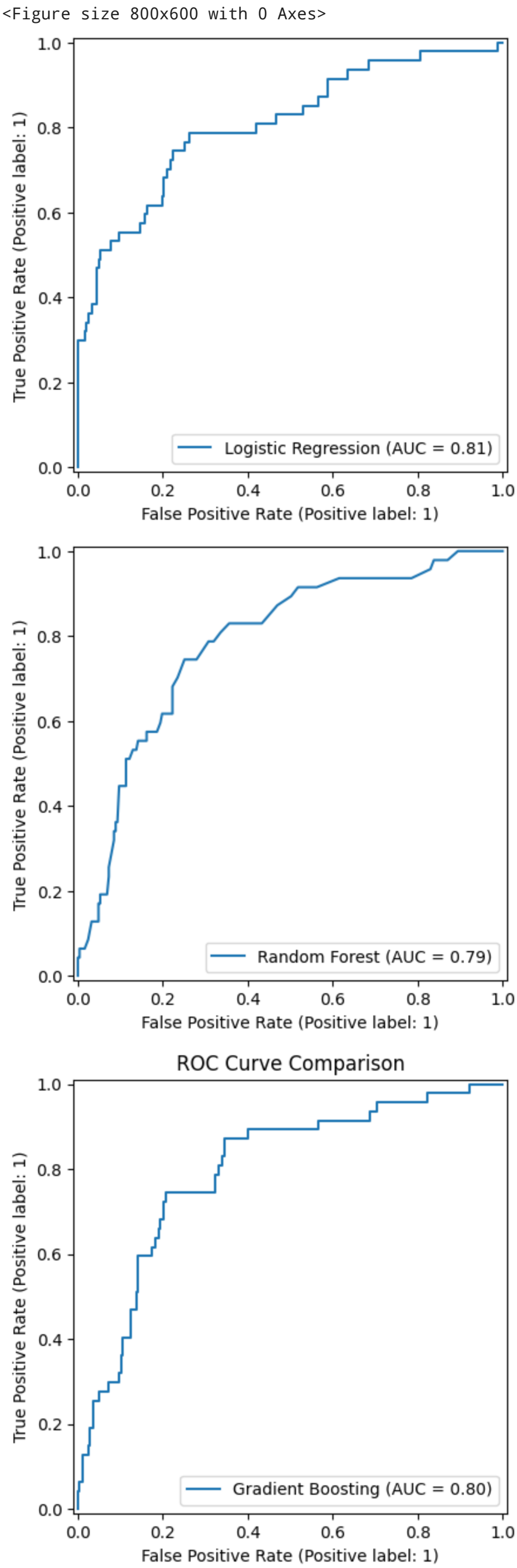
RocCurveDisplay.from_predictions(
    y_test,
    rf_prob,
    name="Random Forest"
)

RocCurveDisplay.from_predictions(
    y_test,
    gb_prob,
    name="Gradient Boosting"
)

plt.title("ROC Curve Comparison")

plt.savefig(
    "charts/roc_curve_comparison.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



Business Insights and Recommendations

Based on the exploratory data analysis and machine learning models, the following key insights and recommendations can help the HR department reduce employee attrition and improve workforce retention.

Key Business Insights